

Bitcoin Market Sentiment vs Trader Performance Analysis

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Role: AI Intern Assignment – PrimeTrade.ai

1. Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using historical trading data from Hyperliquid.

The analysis aims to answer questions such as:

- Does market sentiment influence trader profitability?
 - Do traders use higher leverage during Fear or Greed?
 - Which trading behaviors lead to better performance?
 - Can sentiment be used to improve trading strategies?
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2. Datasets Used

Dataset 1: Bitcoin Fear & Greed Index

Columns

- Date
- Classification (Fear / Greed)

Purpose

Provides the daily market sentiment.

Dataset 2: Historical Hyperliquid Trader Data

Columns include

- Account
- Symbol
- Execution Price
- Size
- Side
- Time
- Leverage
- ClosedPnL
- Start Position
- Event

Purpose

Contains historical trading activity and trader performance.

3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded both datasets using Pandas
 - Converted date columns into datetime format
 - Removed duplicate records
 - Checked missing values
 - Extracted trading date from timestamp
 - Merged trading data with Fear & Greed Index using the Date column
 - Verified data consistency
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4. Exploratory Data Analysis

The following analyses were performed.

4.1 Average Profit During Fear vs Greed

Insert Chart

Observation

Example:

Traders generated a higher average ClosedPnL during Greed periods compared to Fear periods.

4.2 Number of Trades

Insert Chart

Observation

Example

Trading activity increased significantly during Greed periods.

4.3 Profit Distribution

Insert Histogram

Observation

Example

Fear periods showed greater variability in trader profits, indicating increased market volatility.

4.4 Average Leverage

Insert Chart

Observation

Example

Traders used higher leverage during Fear, increasing both risk and reward.

4.5 Buy vs Sell Analysis

Insert Chart

Observation

Example

Buy positions were more common during Greed while short positions increased during Fear.

4.6 Symbol-wise Profitability

Insert Chart

Example:

- BTC
- ETH
- SOL

Observation

Example

BTC generated the highest overall profit among all traded assets.

4.7 Top Performing Traders

Insert Table

Account	Avg PnL	Win Rate
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Observation

Example

Top traders maintained consistent profitability with moderate leverage instead of maximum leverage.

4.8 Correlation Analysis

Insert Heatmap

Observation

Example

ClosedPnL had a moderate positive relationship with trade size but only a weak correlation with leverage.

5. Key Insights

Example

Insight 1

Greed periods contained more trading activity than Fear periods.

Insight 2

Fear periods showed higher market volatility.

Insight 3

Using extremely high leverage did not consistently improve profitability.

Insight 4

Successful traders maintained better risk management rather than simply increasing position size.

6. Business Recommendations

Based on the analysis, the following recommendations can be made.

- Reduce leverage during Fear periods.
 - Increase position sizing only after confirming market trend.
 - Use market sentiment as an additional feature for trading decisions.
 - Monitor trader risk rather than focusing only on profit.
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7. Conclusion

This analysis demonstrates that market sentiment influences trading behavior and trader performance.

Greed periods generally encourage higher trading activity, whereas Fear periods increase volatility and trading risk.

Combining sentiment analysis with trading metrics can help build smarter trading strategies and improve risk management.

8. Future Work

Possible improvements include:

- Build a predictive machine learning model
- Create a real-time dashboard
- Perform sentiment forecasting
- Analyze account-level trading strategies